Network Security

DaeHun Nyang

I am DaeHun Nyang.

- This is my 7th year at Ewha.
- A professor at Inha university for 17+ years
- A researcher at ETRI for 2.5 years
- Education: Degree from Yonsei (MS/PhD), KAIST (BS)

- Service
  - Section editor of ETRI Journal
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Research Focus

- Network packet/flow monitoring for intrusion detection
  - Data Sketch
  - Differential privacy
- Encrypted traffic analysis
  - TOR, TLS
- Efficient AI
  - AI model compression
  - KV-cache compression
- Efficient Storage for AI
  - Log-Structured Indexing

Home page

- http://seclab.ewha.ac.kr

In this class, you will learn and study

- Network protocol architectures
- Cryptographic primitives: Public Key, Digital Signature, MAC, Hash
- Layer 3 Security protocol: IPSEC
- Layer 4 Security protocol: TLS (Transport Layer Security) 1.2 vs. 1.3
- Firewall
- QUIC
- Block chain
- Network Intrusion Detection

Multiple invited talks are planned.

Week 1	2025/03/04(TUE)	Orientation and introduction to network security.
	2025/03/06(THU)	User authentication and Password cracker
Week 2	2025/03/11(TUE)	User authentication
	2025/03/13(THU)	Network protocol architectures
Week 3	2025/03/18(TUE)	Public Key
	2025/03/20(THU)	Public Key
Week 4	2025/03/25(TUE)	Digital Signature
	2025/03/27(THU)	Digital Signature, Hash
Week 5	2025/04/01(TUE)	MAC
	2025/04/03(THU)	MAC
Week 6	2025/04/08(TUE)	IPSec
	2025/04/10(THU)	IPSec
Week 7	2025/04/15(TUE)	TLS 1.2
	2025/04/17(THU)	TLS 1.2
Week 8	2025/04/22(TUE)	Midterm Exam
	2025/04/24(THU)	Midterm Exam

Topics are subject to change.

Week 9	2025/04/29(TUE)	TLS Attack
	2025/05/01(THU)	Labor day
Week 10	2025/05/06(TUE)	Alternative Holiday - Buddha's Birthday
	2025/05/08(THU)	Invited Talk
Week 11	2025/05/13(TUE)	Traffic Analysis
	2025/05/15(THU)	Traffic Analysis
Week 12	2025/05/20(TUE)	Network Intrusion Detection
	2025/05/22(THU)	Network Intrusion Detection
Week 13	2025/05/27(TUE)	Doss Attack
	2025/05/29(THU)	Invited Talk
Week 14	2025/06/03(TUE)	Doss Attack
	2025/06/05(THU)	Doss Attack
Week 15	2025/06/10(TUE)	Firewall
	2025/06/12(THU)	Firewall
Makeup Classes 1	2025/05/01(THU)	Zoom: TLS attack
Makeup Classes 2	2025/05/06(TUE)	Zoom: TLS attack

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Grading policy

- Midterm Exam 40%
- Final Exam 40%
- Homework/Presentation 10%
  - Intrusion detection project
  - Packet traces are given.
  - You must be able to do AI modeling to detect intrusion in the network trace.
- Participation 10%

- Absence
  - Prepare and submit convincing evidence of your non-absence to **TA**. (not to me)
- Tardiness
  - Only on the day.
- TA: Gayoung Kim (김가영 조교) - gayoungkim@ewha.ac.kr

HW: Password cracking

- Password is stored not in a plaintext but in a hashed form.
- Hash function is oneway and you cannot inverse it.
  - Y = Hash(X), where X cannot be found easily from Y.
  - nyang/X -> nyang/Hash(X)
  - The attacker guesses X’ from a dictionary.
  - It can calculate H(X’)
  - Check if H(X’) == H(X)?
  - Dictionary Attack = offline guessing attack
  - Simple, short, not-random and guessable password is not good.
- You are given a file that has a lot of hashed passwords.
- You have to find out passwords as many as possible.
- The hash function will be given.

AI Engineering/Security 분야 인턴, 대학원생 모집

* 모집대상:

  사이버보안학과, 컴퓨터공학과, 수학과 학부생 또는 졸업생

  인공지능대학 사이버보안학과 대학원의

  석사, 박사 또는 석박사통합과정 진학을 희망하는 학생

  알고리즘 프로그래밍 능력 필요

* 연구주제:

  인공지능 engineering, 시스템 engineering, 알고리즘

* 연락처:

  양대헌 교수, nyang@ewha.ac.kr (CV, 성적표 첨부)